

AI Capstone Project Report :

InterviewAI - AI Powered Interview Preparation Assistant

SDG 4 - Quality Education

1. Selected SDG and Reason for Selection

Selected Sustainable Development Goal

SDG 4 – Quality Education

About the SDG

SDG 4 aims to provide quality education and learning opportunities for everyone while helping people develop skills for employment and personal growth.

Reason for Selection

Many students and fresh graduates possess technical knowledge but lack interview preparation and practical communication skills. Access to personalized interview coaching is often expensive and not available to everyone. As a result, many capable candidates struggle during placement drives and job interviews.

This issue directly impacts employability and career opportunities. By providing AI-powered interview practice and feedback, InterviewAI helps students improve their confidence, communication skills, and interview readiness, supporting the goals of quality education and skill development.

2. Problem Statement

Many students and fresh graduates face difficulty during job interviews because they lack sufficient interview practice and personalized feedback.

This problem is common in colleges and universities where students often prepare using static resources and generic interview questions.

As a result, candidates may struggle to communicate effectively, identify weaknesses, and build confidence before actual interviews.

Without proper preparation, students may miss employment opportunities despite having the required technical skills and experience reduced career growth.

3. Proposed Solution

Project Overview

InterviewAI is an AI-powered interview preparation platform designed to help students practice interviews in a realistic environment and receive instant feedback.

How the System Works

1. User selects interview settings such as interview type and experience level.
2. The AI generates interview questions based on the selected criteria.
3. The user answers the questions in text format.
4. The AI evaluates the answer based on multiple metrics.
5. Feedback, scores, and ideal answers are generated.
6. The system continues the interview through multiple rounds.
7. Users can also use Resume-Based Mode by providing resume content for personalized interview questions.

How the Solution Solves the Problem

InterviewAI provides accessible and personalized interview preparation without requiring a human mentor. It allows students to practice repeatedly, receive feedback instantly, and improve weak areas before attending real interviews.

Target Users

- College students
 - Fresh graduates
 - Internship applicants
 - Job seekers
-

4. Project Features

1. General Interview Mode

Users can practice technical, HR, behavioral, and mixed interviews. Questions are generated dynamically by AI based on the selected interview type.

2. Resume-Based Interview Mode

Users can provide resume content, and the AI generates personalized questions based on projects, skills, and technologies mentioned in the resume.

3. Adaptive Difficulty Levels

The system supports Fresher, Intermediate, and Experienced levels. Question complexity changes according to the selected experience level.

4. Multi-Round Interview Simulation

InterviewAI conducts interviews across multiple rounds, creating a more realistic interview experience compared to single-question systems.

5. AI-Powered Evaluation

Every answer is evaluated by AI using multiple criteria including knowledge, clarity, and confidence.

6. Knowledge Score

The system evaluates the technical correctness and completeness of the candidate's answer.

7. Clarity Score

The AI measures how clearly and effectively the answer is communicated.

8. Confidence Score

The system estimates how confidently the response is presented.

9. Ideal Answer Generation

For every question, the AI provides a model answer that helps users understand how an ideal response should be structured.

10. Company and Role Specific Preparation

Users can specify their target role and organization to receive more relevant interview questions.

5. Technology Stack

Frontend

- Next.js
- TypeScript
- Tailwind CSS
- shadcn/ui

AI

- Groq API
- Llama 3.3 70B Model

Tools

- Git
- GitHub

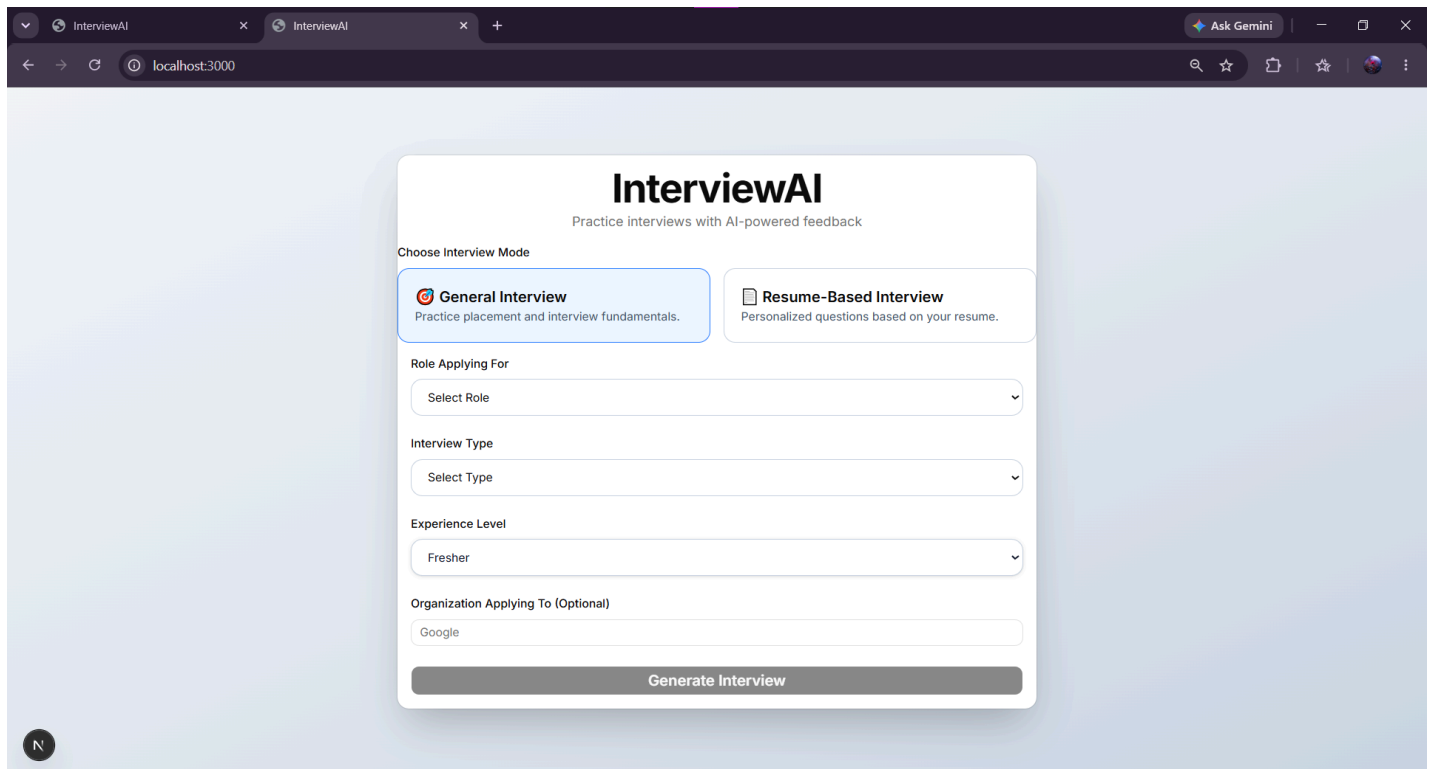
- VS Code

Deployment:

- Vercel

6. Project Screenshots

1. Home Page



The screenshot shows a web browser window with two tabs labeled 'InterviewAI'. The address bar shows 'localhost:3000'. The page features a central white card with the title 'InterviewAI' and the subtitle 'Practice interviews with AI-powered feedback'. Below the title, there are two buttons for 'Choose Interview Mode': 'General Interview' (highlighted in blue) and 'Resume-Based Interview'. The 'General Interview' button has a subtext 'Practice placement and interview fundamentals.' and the 'Resume-Based Interview' button has a subtext 'Personalized questions based on your resume.' Below these buttons are four dropdown menus: 'Role Applying For' (with 'Select Role' as the placeholder), 'Interview Type' (with 'Select Type' as the placeholder), 'Experience Level' (with 'Fresher' as the placeholder), and 'Organization Applying To (Optional)' (with 'Google' as the placeholder). At the bottom of the card is a large grey button labeled 'Generate Interview'. A small circular logo with a stylized 'N' is visible in the bottom left corner of the page.

2. Interview Question Generation

The screenshot shows the InterviewAI web application in a browser window. The page has a dark header with the title 'InterviewAI' and the subtitle 'Practice interviews with AI-powered feedback'. Below the header, there are two tabs: 'InterviewAI' and 'Ask Gemini'. The main content area is white and contains a form for generating interview questions. The form has two sections: 'General Interview' and 'Resume-Based Interview'. The 'General Interview' section is selected and contains the following fields: 'Role Applying For' (Frontend Developer), 'Interview Type' (HR), 'Experience Level' (Fresher), and 'Organization Applying To (Optional)' (Google). Below these fields is a 'Generate Interview' button. The 'Resume-Based Interview' section is also visible but not selected. Below the 'Generate Interview' button, there is a section for 'Question 1 / 5' with the question 'What is the basic difference between HTML and CSS?'. Below the question is a text input field containing the answer 'html is skeleton and css is apperance'. Below the input field is an 'Evaluate Answer' button.

InterviewAI

Practice interviews with AI-powered feedback

Choose Interview Mode

General Interview
Practice placement and interview fundamentals.

Resume-Based Interview
Personalized questions based on your resume.

Role Applying For
Frontend Developer

Interview Type
HR

Experience Level
Fresher

Organization Applying To (Optional)
Google

Generate Interview

Question 1 / 5
What is the basic difference between HTML and CSS?

html is skeleton and css is apperance

Evaluate Answer

3. Answer Submission and AI Evaluation

The screenshot shows the InterviewAI web application in a browser window, displaying the AI evaluation interface. The page has a dark header with the title 'InterviewAI' and the subtitle 'Practice interviews with AI-powered feedback'. Below the header, there are two tabs: 'InterviewAI' and 'Ask Gemini'. The main content area is white and contains a form for evaluating the answer. The form has two sections: 'General Interview' and 'Resume-Based Interview'. The 'General Interview' section is selected and contains the following fields: 'Role Applying For' (Frontend Developer), 'Interview Type' (HR), 'Experience Level' (Fresher), and 'Organization Applying To (Optional)' (Google). Below these fields is a 'Generate Interview' button. The 'Resume-Based Interview' section is also visible but not selected. Below the 'Generate Interview' button, there is a section for 'Question 1 / 5' with the question 'What is the basic difference between HTML and CSS?'. Below the question is a text input field containing the answer 'html is skeleton and css is apperance'. Below the input field is an 'Evaluate Answer' button. Below the 'Evaluate Answer' button, there is a section for 'AI Feedback' with the following text: 'Knowledge Score: 6/10', 'Clarity Score: 8/10', 'Confidence Score: 7/10', 'Strength: The candidate has grasped the basic concept of HTML and CSS.', 'Improvement: The candidate should provide more detailed and formal explanations, such as "HTML is the structure and CSS is the styling".', and 'Ideal Answer: HTML is the skeleton, CSS is the appearance, HTML provides structure, and CSS provides styling.' Below the feedback section is a 'Next Question' button.

InterviewAI

Practice interviews with AI-powered feedback

Choose Interview Mode

General Interview
Practice placement and interview fundamentals.

Resume-Based Interview
Personalized questions based on your resume.

Role Applying For
Frontend Developer

Interview Type
HR

Experience Level
Fresher

Organization Applying To (Optional)
Google

Generate Interview

Question 1 / 5
What is the basic difference between HTML and CSS?

html is skeleton and css is apperance

Evaluate Answer

AI Feedback

Knowledge Score: 6/10
Clarity Score: 8/10
Confidence Score: 7/10

Strength: The candidate has grasped the basic concept of HTML and CSS.

Improvement: The candidate should provide more detailed and formal explanations, such as "HTML is the structure and CSS is the styling".

Ideal Answer: HTML is the skeleton, CSS is the appearance, HTML provides structure, and CSS provides styling.

Next Question

InterviewAI

Ask Gemini

localhost:3000

Experience Level

Fresher

Organization Applying To (Optional)

Google

Generate Interview

Question 5 / 5

What is the role of a tag in HTML?

it gives ability to apply different structures for different purpose

Evaluate Answer

AI Feedback

Knowledge Score: 6/10

Clarity Score: 5/10

Confidence Score: 4/10

Strength: The candidate attempts to relate tags to applying structures for different purposes.

Improvement: The candidate should provide more specific and accurate information about the role of tags in HTML.

Ideal Answer:

A tag in HTML is used to define an element.

It provides a way to structure and organize content.

HTML tags are surrounded by angle brackets and usually come in pairs.

Interview Complete 🎉

You completed all 5 interview rounds.

4. Resume-Based Interview Mode

InterviewAI

Ask Gemini

localhost:3000

InterviewAI

Practice interviews with AI-powered feedback

Choose Interview Mode

General Interview

Practice placement and interview fundamentals.

Resume-Based Interview

Personalized questions based on your resume.

Role Applying For

Full Stack Developer

Interview Type

Technical

Experience Level

Fresher

Organization Applying To (Optional)

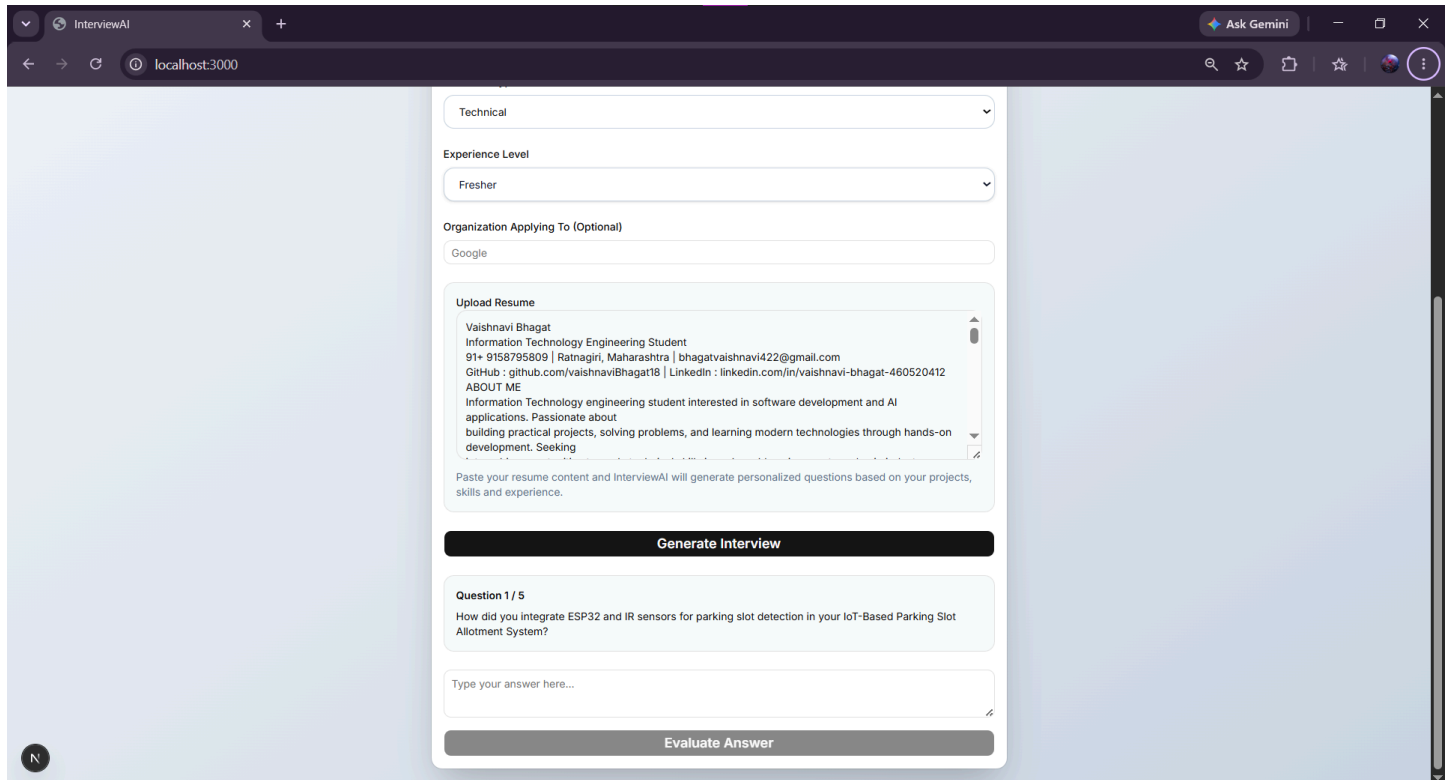
Google

Upload Resume

Vaishnavi Bhagat
Information Technology Engineering Student
91+ 9158795809 | Ratnagiri, Maharashtra | bhagatvaishnavi422@gmail.com
GitHub : github.com/vaishnaviBhagat18 | LinkedIn : linkedin.com/in/vaishnavi-bhagat-460520412
ABOUT ME
Information Technology engineering student interested in software development and AI applications. Passionate about building practical projects, solving problems, and learning modern technologies through hands-on development. Seeking

Paste your resume content and InterviewAI will generate personalized questions based on your projects, skills and experience.

5. Personalized Resume Question



The screenshot shows a web browser window with the address bar displaying 'localhost:3000'. The page title is 'InterviewAI'. The interface includes a sidebar with a search icon and a main content area. The main content area contains a form with the following sections:

- Technical**: A dropdown menu with 'Technical' selected.
- Experience Level**: A dropdown menu with 'Fresher' selected.
- Organization Applying To (Optional)**: A text input field containing 'Google'.
- Upload Resume**: A text area containing a resume for 'Vaishnavi Bhagat', an Information Technology Engineering Student. The resume includes contact information (91+ 9158795809, Ratnagiri, Maharashtra, bhagatvaishnavi422@gmail.com), GitHub profile (github.com/vaishnavibhagat18), LinkedIn profile (linkedin.com/in/vaishnavi-bhagat-460520412), and a brief 'ABOUT ME' section describing her as an IT engineering student interested in software development and AI applications. Below the text area is a note: 'Paste your resume content and InterviewAI will generate personalized questions based on your projects, skills and experience.'
- Generate Interview**: A black button.
- Question 1 / 5**: A section showing a question: 'How did you integrate ESP32 and IR sensors for parking slot detection in your IoT-Based Parking Slot Allotment System?'. Below the question is a text input field with the placeholder 'Type your answer here...'. At the bottom of this section is a grey button labeled 'Evaluate Answer'.

7. Open Source Repository

GitHub Repository: <https://github.com/vaishnaviBhagat18/InterviewAI.git>

8. Future Scope

- PDF Resume Parsing
 - Voice-Based Interviews
 - Progress Tracking Dashboard
 - Company-Specific Interview Preparation
 - Mobile Application
 - Advanced Performance Analytics
-

Conclusion

InterviewAI is an AI-powered interview preparation platform that helps students and job seekers improve confidence, communication skills, and interview readiness through practice and instant feedback. The project supports SDG 4 – Quality Education by providing accessible and personalized learning opportunities for career development.